

Question Paper

Exam Date & Time: 28-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIRST SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025
APPLIED CHEMISTRY FOR ENGINEERS [CHM-1003]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) (i) Explain the construction of polymer membrane electrolyte fuel cells (PEMFC) with a neat-labeled diagram. Write anodic and cathodic reaction for PEMFC. **3M** (4)
A) (ii) Write the discharging anodic and cathodic reactions for lead acid batteries. **1M**

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) Calculate the E.M.F. of the zinc-silver cell at 25°C when (4)
 $[Zn^{2+}] = 0.002\text{ M}$ and $[Ag^+] = 3.0\text{ M}$. Write the cell representation and cell reactions. Calculate ΔG for the cell reaction.

Course Outcome: "CO1" / Cognitive Level: "Application"

- C) Give reason: (2)
(i) Proton conducting membranes provides excellent resistance gas crossover. **1M**
(ii) Li-ion batteries do not contain lithium metal. **1M**

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2) (i) Explain the basic structure of vertically aligned nematic (VAN) displays with a suitable diagram and mention two advantages and disadvantages. **3M** (4)
A) (ii) Mention two of its applications. **1M**

Course Outcome: "CO2" / Cognitive Level: "Application"

- B) (i) Calculate the number average, weight average molecular weight and PDI of polymer molecules with different degrees of polymerization such as 500, 600, 450 and 750 that are mixed in a molecular ratio 1: 2: 3: 4 in a sample of high polymer of vinyl chloride (molecular mass = 62.5). **3M** (4)
(ii) Give reason: Polyvinyl chloride has less slipping power than polyethylene. **1M**

Course Outcome: "CO2" / Cognitive Level: "Application"

- C) (i) Define relative and intrinsic viscosity? **1M** (2)
(ii) As per the data obtained from GPC instrument, the retention time for a polymer sample is 17.0 min. Calculate the molecular weight of the polymer sample (Given: Slope = -0.3 and intercept = 7). **1M**

Course Outcome: "CO2" / Cognitive Level: "Understand"

- 3) (i) Define and explain amperometric and piezoelectric transducers in biosensors with their mechanism. **2M** (4)
- A) (ii) Give reason: Ion-selective electrodes are used in water quality monitoring. **1M**
- (iii) The absorbance of a 2 mL aliquot of the diluted solution is measured as 0.4. Given the calibration graph slope is 0.9 absorbance units per mg of copper, calculate the concentration of copper in the diluted solution. **1M**

Course Outcome: "CO3" / Cognitive Level: "Understand"

- B) (i) What is a catalytic bead sensor? Explain about the three main components of catalytic bead sensors. **2M** (4)
- (ii) Which are the factors affecting the potential of an electrode and mention any two advantages of potentiometric titration. **2M**

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- C) (i) Draw a neat-labelled diagram of Gas sensors. **1M** (2)
- (ii) Calculate the actual concentration of an unknown sample given the Flame photometry instrument reading of 8.15 ppm and dilution factor of 0.01. **1M**

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 4) (I) List the potential health effects of mercury and cadmium exposure. **1M** (4)
- A) (II) Explain the circular economy model in E-waste management through (a) Design for longevity and repairability and (b) Reuse and refurbishment. **2M**
- (III) Describe any two steps to achieve a circular economy in the Electronics sector. **1M**

Course Outcome: "CO4" / Cognitive Level: "Application"

- B) (I) Explain in detail the four disposal challenges associated with PFAS. **2M** (4)
- (II) Describe air pollution through improper e-waste disposal. **2M**

Course Outcome: "CO4" / Cognitive Level: "Understand"

- C) Describe the chemical reactions that occur in a lead-acid battery during charging and overcharging. (2)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 5) (i) Explain the principle of colorimetric analysis and mention the name of sample cell used in measuring absorbance? **(2M)** (4)
- A) (ii) A flame photometer produces an intensity of 100 units for a 1.5 mg/L solution. What intensity should be observed for 2 mg/L? **2M**

Course Outcome: "CO3" / Cognitive Level: "Understand"

- B) (i) Classify biodegradable polymers, mention any two advantages and applications. **3M** (4)
- (ii) Discuss nanomaterials based on origin with examples. **1M**

Course Outcome: "CO2" / Cognitive Level: "Application"

- C) Explain in detail the preparation of nanomaterials by ball-milling method. (2)

Course Outcome: "CO2" / Cognitive Level: "Understand"

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Event Name	Subject
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#	Course Outcome	Description	Marks
1	1	CO 1	12
2	2	CO 2	16
3	3	CO 3	14
4	4	CO 4	8
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	7	20(40%)
3	Application	7	26(52%)
4	Analysis	1	4(8%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)